

# Answer to Practice Problems of Exam 2.

1. (a)  $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n \sin n}{n^2 + 1}$   
 $= \lim_{n \rightarrow \infty} \frac{\sin n}{n + 1/n} \quad (|\sin n| \leq 1)$   
 $= 0 \quad (\text{by the squeezing theorem}) \rightarrow \text{Convergent}$

(b)  $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left(1 + \frac{3}{n}\right)^n$   
 $\stackrel{n=3m}{=} \lim_{m \rightarrow \infty} \left(1 + \frac{1}{m}\right)^{3m}$   
 $= \left(\lim_{m \rightarrow \infty} \left(1 + \frac{1}{m}\right)^m\right)^3$   
 $= e^3 \rightarrow \text{Convergent}$

2. (a)  $\lim_{n \rightarrow \infty} \ln\left(\frac{n}{3n+1}\right) = \ln\left(\frac{1}{3}\right) \neq 0$   
 $\Rightarrow \sum_{n=1}^{\infty} \ln\left(\frac{n}{3n+1}\right) \text{ diverges.}$

(b) Use Comparison Test  
 let  $b_n = \frac{n}{\sqrt{n^5}} = \frac{1}{n^{3/2}}$   
 clear  $a_n = \frac{n}{\sqrt{n^5 + n + 1}} < \frac{1}{n^{3/2}} = b_n$   
 and  $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$  Converges  
 $\Rightarrow \sum_{n=1}^{\infty} \frac{n}{\sqrt{n^5 + n + 1}}$  Converges.

(c) Use Integration Test  
 let  $f(x) = \frac{1}{x \ln x}$ .  
 decreasing function in  $[2, +\infty)$ .  
 $\int_2^{\infty} f(x) dx = \int_2^{\infty} \frac{1}{x \ln x} dx$   
 $= \lim_{t \rightarrow \infty} \left[ 2 \ln x \right]_{x=2}^{x=t}$   
 $= \lim_{t \rightarrow \infty} 2 [\ln t - \ln 2]$   
 $= +\infty$   
 $\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n \ln n}$  diverges.

(d) Use the Root Test

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{n^{2n}}{(1+2n^2)^n}}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2}{1+2n^2} = \lim_{n \rightarrow \infty} \frac{1}{\frac{1}{n^2} + 2} = \frac{1}{2}$$

$\Rightarrow$  Convergent.

(e)  $\left\{ \begin{array}{l} 0 \leq \frac{6s^2 n}{n^{4/3}} \leq \frac{1}{n^{1/3}} \text{ Since } 0 \leq 6s^2 n \leq 1 \\ \sum_{n=1}^{\infty} \frac{1}{n^{4/3}} \text{ Converges} \end{array} \right.$

$\Rightarrow \sum_{n=1}^{\infty} \frac{6s^2 n}{n^{4/3}}$  Converges by the Comparison Test.

(f)  $\sum_{n=1}^{\infty} n e^{-n} = \sum_{n=1}^{\infty} \frac{n}{e^n}$

Use the Ratio Test.

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)/e^{n+1}}{n/e^n} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{(n+1)e^n}{n e^{n+1}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{(n+1)}{n e} \right| = \frac{1}{e} < 1$$

$$\Rightarrow \sum_{n=1}^{\infty} n e^{-n} \text{ Converges.}$$

(g) Use the Ratio Test.

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{3^{(n+1)} (n+1)^2 / (n+1)!}{3^n n^2 / n!} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{3^{n+1} (n+1)^2 n!}{3^n n^2 (n+1)!} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{3 (n+1)^2}{(n+1) n^2} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{3(n+1)}{n^2} \right| = 0$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{3^n n^2}{n!} \text{ Converges}$$

(h).

$$\frac{n^n}{n!} = \frac{n \cdot n \cdot \dots \cdot n}{1 \cdot 2 \cdot \dots \cdot n} \geq 1.$$

$$\Rightarrow \lim_{n \rightarrow \infty} (-1)^{n-1} \frac{n^n}{n!} \text{ doesn't exist.}$$

$$\Rightarrow \sum_{n=1}^{\infty} (-1)^{n-1} \frac{n^n}{n!} \text{ diverges.}$$

$$3. (a) a_n = \sum_{n=1}^{+\infty} (-1)^{n-1} \frac{1}{n^3}, |a_n| = \frac{1}{n^3}$$

$$\sum_{n=1}^{+\infty} |a_n| = \sum_{n=1}^{+\infty} \frac{1}{n^3} \text{ Converges}$$

$$\Rightarrow \sum_{n=1}^{+\infty} (-1)^{n-1} \frac{1}{n^3} \text{ absolutely converges.}$$

$$(b) a_n = (-1)^n \sin\left(\frac{\pi}{n}\right), |a_n| = \sin\left(\frac{\pi}{n}\right).$$

$$\sum_{n=1}^{+\infty} |a_n| = \sum_{n=1}^{+\infty} \sin\left(\frac{\pi}{n}\right) \quad \text{since } \sin\left(\frac{\pi}{n}\right) > 0 \text{ for } n \geq 1.$$

$$\text{let } b_n = \frac{1}{n}.$$

Use the Limit Comparison Test.

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{|a_n|}{b_n} &= \lim_{n \rightarrow \infty} \frac{\sin\left(\frac{\pi}{n}\right)}{\frac{1}{n}} = \lim_{x \rightarrow 0} \frac{\sin(x)}{x} \\ &= \lim_{x \rightarrow 0} \frac{x \cos(x)}{1} = x \neq 0 \end{aligned}$$

Together with  $\sum_{n=1}^{+\infty} \frac{1}{n}$  diverging, we know that

$$\sum |a_n| = \sum \sin\left(\frac{\pi}{n}\right) \text{ diverges.}$$

On the other hand.

$$(1) \sin\left(\frac{\pi}{n}\right) > \sin\left(\frac{\pi}{n+1}\right)$$

$$(2) \lim_{n \rightarrow \infty} \sin\left(\frac{\pi}{n}\right) = 0$$

$$\Rightarrow \sum_{n=1}^{+\infty} a_n = \sum_{n=1}^{+\infty} (-1)^n \sin\left(\frac{\pi}{n}\right) \text{ converges}$$

by the Alternating Series Test.

$$\Rightarrow \sum_{n=1}^{+\infty} (-1)^n \sin\left(\frac{\pi}{n}\right) \text{ conditionally converges.}$$

$$(c) a_n = \frac{\sin 2n}{1+2^n}$$

$$|a_n| = \frac{|\sin 2n|}{1+2^n} \leq \frac{1}{1+2^n} \leq \frac{1}{2^n}$$

$$\sum_{n=1}^{+\infty} \frac{1}{2^n} \text{ converges}$$

$$\Rightarrow \sum_{n=1}^{+\infty} \frac{|\sin 2n|}{1+2^n} \text{ converges}$$

$$\Rightarrow \sum \frac{\sin 2n}{1+2^n} \text{ absolutely converges.}$$

$$(d) a_n = (-1)^n \frac{\ln n}{n}$$

$$|a_n| = \frac{\ln n}{n} > \frac{1}{n} \text{ when } n \geq 3.$$

$$\sum_{n=1}^{+\infty} \frac{1}{n} \text{ diverges, thus } \sum_{n=2}^{+\infty} |a_n| = \sum_{n=1}^{+\infty} \frac{\ln n}{n}$$

diverges by the Comparison Test.

On the other hand.

$$(a) \frac{\ln n}{n} > \frac{\ln(n+1)}{n+1}, \text{ for } n \geq 3.$$

$$(b) \lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0 \Rightarrow \left(\frac{\ln x}{x}\right)' = \frac{1 - \ln x}{x^2} < 0 \text{ when } x \geq 3.$$

$$\Rightarrow \sum_{n=2}^{+\infty} a_n = \sum_{n=2}^{+\infty} (-1)^n \frac{\ln n}{n} \text{ converges}$$

$$\text{Thus } \sum_{n=1}^{+\infty} (-1)^n \frac{\ln n}{n} \text{ conditionally converges.}$$

$$4. (a) \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\sqrt{n+1} x^{n+1}}{\sqrt{n} x^n} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \sqrt{1 + \frac{1}{n}} x \right| = |x|$$

$$|x| < 1 \Rightarrow R = 1, \text{ i.e. } -1 < x < 1.$$

$$\text{when } x = 1, \sum_{n=1}^{+\infty} \sqrt{n} \text{ diverges}$$

$$x = -1, \sum_{n=1}^{+\infty} (-1)^n \sqrt{n} \text{ diverges}$$

Thus the interval of convergence is  $(-1, 1)$ .

$$(b) \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} x^{n+1} / (n+1)^{2n+1}}{(-1)^n x^n / n^{2n}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} x^{n+1} n^{2n}}{(-1)^n x^n (n+1)^{2n+1}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{x}{2} \frac{n^2}{(n+1)^2} \right| = \lim_{n \rightarrow \infty} \left| \frac{x}{2} \left(\frac{n}{n+1}\right)^2 \right| = \left| \frac{x}{2} \right|$$

$$\left| \frac{x}{2} \right| < 1 \Rightarrow |x| < 2 \Rightarrow R = 2, \text{ i.e. } -2 < x < 2$$

$$\text{When } x = 2, \sum_{n=1}^{+\infty} (-1)^n \frac{2^n}{n^2 2^n} = \sum_{n=1}^{+\infty} (-1)^n \frac{1}{n^2} \text{ converges.}$$

$$x = -2, \sum_{n=1}^{+\infty} (-1)^n \frac{(-2)^n}{n^2 2^n} = \sum_{n=1}^{+\infty} \frac{1}{n^2} \text{ converges.}$$

Thus the interval of convergence is  $[-2, 2]$

$$(c) \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{3^{n+1} (x-3)^{n+1} / \sqrt{n+3}}{3^n (x-3)^n / \sqrt{n}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{3^{n+1} (x-3)^{n+1} \sqrt{n}}{3^n (x-3)^n \sqrt{n+3}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| 3(x-3) \sqrt{\frac{n}{n+3}} \right| = 3|x-3|$$

$$3|x-3| < 1 \Rightarrow |x-3| < \frac{1}{3} \Rightarrow R = \frac{1}{3}$$

$$\text{i.e. } -\frac{1}{3} < x-3 < \frac{1}{3}$$

$$\Rightarrow \frac{8}{3} < x < \frac{10}{3}$$

$$\text{When } x = \frac{8}{3}, \sum_{n=1}^{+\infty} \frac{3^n (\frac{8}{3} - 3)^n}{\sqrt{n+3}} = \sum_{n=1}^{+\infty} \frac{3^n \cdot (-\frac{1}{3})^n}{\sqrt{n+3}}$$

$$= \sum_{n=1}^{+\infty} (-1)^n \frac{1}{\sqrt{n+3}} \text{ Converges}$$

$$x = \frac{10}{3}, \sum_{n=1}^{+\infty} \frac{3^n (\frac{10}{3} - 3)^n}{\sqrt{n+3}} = \sum_{n=1}^{+\infty} \frac{3^n (\frac{1}{3})^n}{\sqrt{n+3}}$$

$$= \sum_{n=1}^{+\infty} \frac{1}{\sqrt{n+3}} \text{ diverges}$$

Thus, the interval of convergence is

$$\left[ \frac{8}{3}, \frac{10}{3} \right)$$

$$5. (a) \frac{1}{1+x} = \frac{1}{1-(-x)} = \sum_{n=0}^{+\infty} (-x)^n = \sum_{n=0}^{+\infty} (-1)^n x^n$$

$$\Rightarrow \frac{x^2}{1+x} = x^2 \cdot \left( \frac{1}{1+x} \right) = x^2 \sum_{n=0}^{+\infty} (-1)^n x^n$$

$$= \sum_{n=0}^{+\infty} (-1)^n x^{n+2}$$

$$\lim_{n \rightarrow +\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow +\infty} \left| \frac{(-1)^{n+1} x^{n+3}}{(-1)^n x^{n+2}} \right|$$

$$= |x|$$

$$\Rightarrow R = 1.$$

$$(b) e^{2x} = \sum_{n=0}^{+\infty} \frac{(2x)^n}{n!} = \sum_{n=0}^{+\infty} 2^n \frac{x^n}{n!}$$

$$x e^{2x} = x \sum_{n=0}^{+\infty} 2^n \frac{x^n}{n!}$$

$$= \sum_{n=0}^{+\infty} \frac{2^n x^{n+1}}{n!}$$

$$\lim_{n \rightarrow +\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow +\infty} \left| \frac{2^{n+1} x^{n+2} / (n+1)!}{2^n x^{n+1} / n!} \right|$$

$$= \lim_{n \rightarrow +\infty} \left| \frac{2x}{n+1} \right| = 0$$

$$\Rightarrow R = +\infty$$

$$(c) \sin(x^2) = \sum_{n=0}^{+\infty} \frac{(-1)^n (x^2)^{2n+1}}{(2n+1)!}$$

$$= \sum_{n=1}^{+\infty} \frac{(-1)^n x^{4n+2}}{(2n+1)!}$$

$$\lim_{n \rightarrow +\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow +\infty} \left| \frac{(-1)^{n+1} x^{4(n+1)+2} / (2(n+1)+1)!}{(-1)^n x^{4n+2} / (2n+1)!} \right|$$

$$= \lim_{n \rightarrow +\infty} \left| \frac{x^4}{(2n+2)(2n+3)} \right| = 0$$

$$\Rightarrow R = +\infty$$